

MOVE VALUE BASED CHESS ENGINE

ICS3U CULMINATING DRAFT DOCUMENTATION

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Requirements:**Dictionary of specific terminology**

Term	Definition
Chess Board	An 8 x 8 grid where the game will be played. The board will be displayed in a Tkinter window that is 700x700 pixels.
Chess Piece	Game pieces like a pawn, rook, knight, bishop, and etc.
Player	User playing the game. Opposing the AI Opponent.
AI Opponent	The algorithmic computer controlled opponent that will be the opposing user.
Legal Move	A move that follows the rules of chess.
Check	When the King is under the threat of being captured.
Checkmate	When the King has no legal move to escape and is under the threat of being captured.
Stalemate	When a player has no more legal moves, but their king is not currently in check, resulting in a draw.
Point Value	Point value assigned to each piece. For example, both the knight and bishop are worth 3 points. Where the queen is worth 9 points.
Point-Based AI	The AI opponent chooses moves by maximizing move value instead of using the generalized min-max algorithm.
Game Over	The condition where the game ends because of either a checkmate or a point-based win from the AI opponent. Another instance of the game ending could be running out of time.
Timer	A count down from a specific time that will indicate when the game ends (winner will be based on who has the highest amount of points).
Position Value	The score given to a move based on how

	useful the destination square is.
Development Value	The score given to a move that helps move pieces from their starting positions into more useful positions. For example, moving the knight to the centre to gain more control.
Safety Value	The score given to a move based on how safe the moved piece is after the move. A move has a lower safety value if it places an important piece where it can be captured.
Small Random Tie-Break Value	A small random score added to a move when multiple legal moves account to similar values. This is to prevent the AI from making the same moves each time. Value must be really small so that it does not deter it from its algorithmic behaviour.
Move Value	The total score assigned to a possible AI move. It is calculated by combining the point value, position value, development value, safety value, and a small random tie-break value. The AI will use the move value to determine the best legal move.
Random Promotion	The twist: at a random point in the game, one AI pawn will be promoted to a stronger piece. Contingent on move value analysis, the promotion can alter inner variables like position value, safety value, and development value, making the promoted piece more impactful.
Bias Value	A value that is derived from the predicted player's move value after the chosen AI move. This value can only be subtracted from the total move value as a way to prevent risky moves.

User Standpoint Description

For the user playing against the AI opponent, the software must provide the following:

Game Setup

- The game launches and displays a chess board.

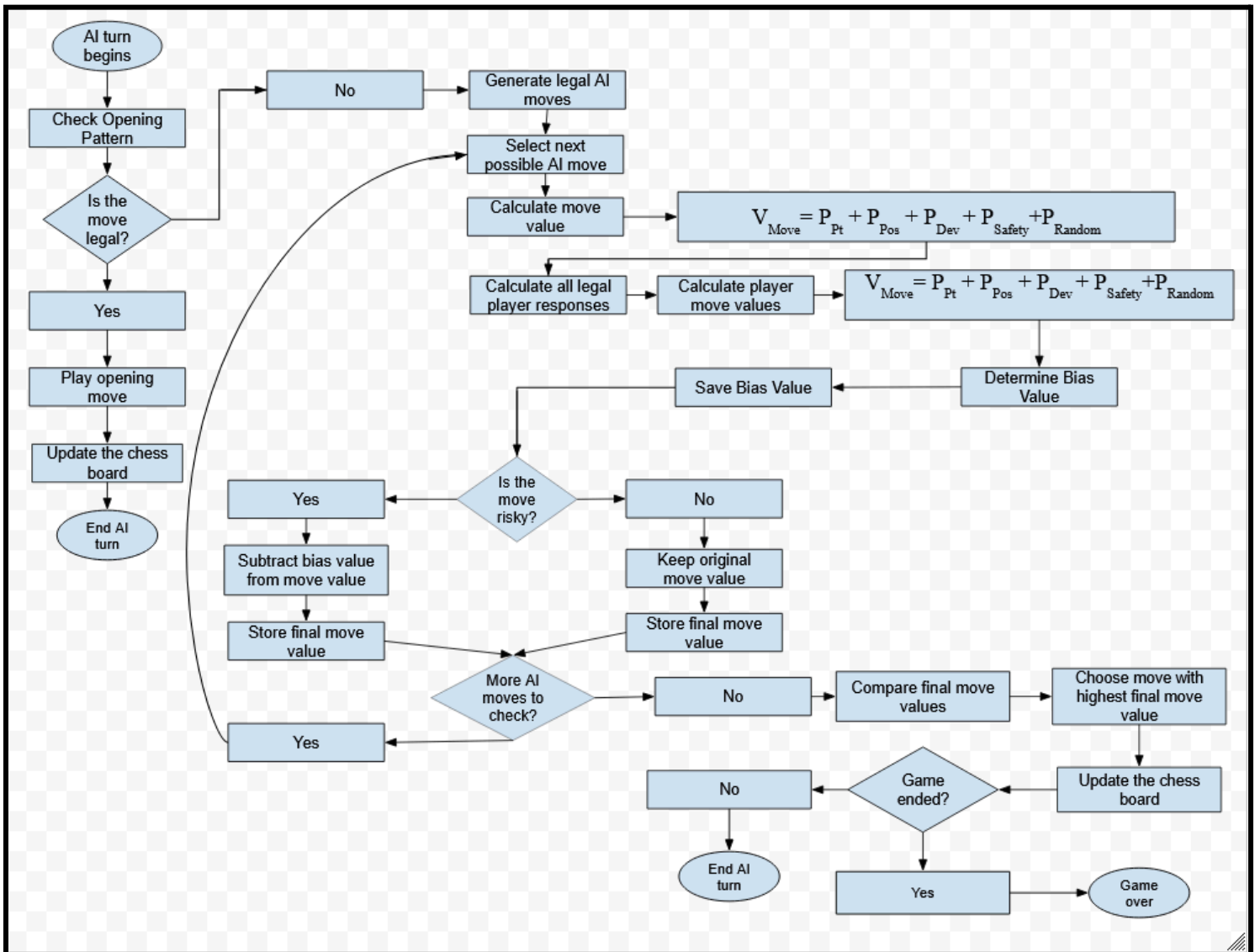
- All chess pieces will be in their starting positions.
- The player will be allotted the bottom half of the board.
- The AI will control the opposite side of the board.
- Either side of the board will be based on a randomized possibility of black or white.
- A timer will be displayed and will begin counting down after the first move.
- Both the player and AI will have their own timers.
- The user can see all the chess board, chess pieces and their placement, current score, and the time remaining.

Gameplay

- The user can select a chess piece and attempt to move it.
- The engine will verify whether it is legal or not
 - If the move is illegal the move will not be allowed and the canvas will not update and will remain in its initial position.
 - If the move is legal then the canvas will update to the player's move.
- After the player makes a legal move the AI opponent will take its turn.
- The AI opponent chooses a move by scoring possible legal moves using point values, position value, development value, and safety value.
- At the beginning of the game, the AI will follow a randomly selected opening pattern.
- If the current opening pattern cannot be used because of the player's moves, the AI will switch to its general point-based move system.
- The AI will generate all possible legal moves for its own pieces.
- Each possible move will receive a move value based on the point value, position value, development value, safety value, and a small random tie-break value.
- The AI will determine the player's possible moves after each possible AI move and use the same move value system to attach values. Based on each possibility, the AI will subtract a bias value from risky moves to lower the move value to prevent playing risky moves.
- The move with the highest move value will be played by the AI.
- After the AI's move, the canvas will update and it will once again be the player's turn.
- If a piece is captured, the captured piece will be removed from the board.
 - The point value assigned to the piece will be allotted to either the player or AI depending on who captured the piece.
- The game will continue until checkmate, stalemate, the timer running out, or the AI reaching the required point total.

Detailed Requirement:

The following is a flowchart of how the point-based move system works:



Test Case:

Test Case 01	Description
AI Risky Move Test	The AI has multiple possible legal moves. One move gives a higher capture, but allows the player to capture an important AI piece next turn. The AI calculates the player's response and subtracts the bias value and compares all move values. The AI should choose the move with the highest final move value.